Limitations

While our experimental setup spans 13 models of different types and sizes, and explores different sampling temperature values, the scale of the prompts in our test suite remains limited due to the difficulty of manually creating prompts that are likely to leak in a way that we can detect and evaluate. This is also often the case when measuring other language model biases. In addition, though the results are consistent across all models and languages we experiment with, the trends might be different with other models or languages we have not tested.

Finally, we cannot guarantee that the automatic evaluations do not include noise: in some cases, even after our automatic post-processing of the generations, our automatic metrics might consider non-leaking instance as leaking (e.g., in cases of occurrences of the concept in the generation due to a partial repetition of the prompt). However, the human evaluation we conduct generally agrees with the findings we get with the automatic detection, supporting their reliability.

Ethical Considerations

Semantic leakage is a broad umbrella that may encompass many types of biases, and it broadens our perception and understanding about association biases that are being learned by models and then exhibited in generation. As such, semantic leakage might manifest at times as a concrete type of bias, e.g., cultural, gender or racial biases. This depends on the specific setting and the chosen concept that leaks; while our prompts and concepts are fairly general, these biases and stereotypes could potentially surface in our evaluation.[19]

However, our main goal in this paper is not to focus on specific types of biases (or biases that are particularly harmful) but to characterize and quantify the broader class of semantic leakage. We believe that a better understanding of the greater picture will in turn arm us with better tools to combat the subset of biases and behaviors that do have harmful consequences in practice.

Apart from that, in this paper the annotations and translations were done by several volunteers that were willing to help for the project. The load of this work was very low.

[19]For example, some concepts in the evaluation involve food and music associated with specific countries or ethnicities.

Acknowledgements

This research received support through Schmidt Sciences, LLC. We thank Shauli Ravfogel and Ido Levin for helpful discussions and ideas. We also thank the translators and annotators for their contribution to the paper (Weijia Shi, Jacqueline He, Jacob Schreiber) and the ARK lab members for their valuable feedback. Finally, we would like to express our gratitude to the reviewers of this paper for their useful comments and suggestions.

References

Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*.

Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, et al. 2023. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*.

Giuseppe Attanasio, Debora Nozza, Dirk Hovy, and Elena Baralis. 2022. Entropy-based attention regularization frees unintended bias mitigation from lists. *arXiv preprint arXiv:2203.09192*.

Tolga Bolukbasi, Kai-Wei Chang, James Y Zou, Venkatesh Saligrama, and Adam T Kalai. 2016. Man is to computer programmer as woman is to homemaker? debiasing word embeddings. In *Advances in Neural Information Processing Systems*.

Aylin Caliskan, Joanna J Bryson, and Arvind Narayanan. 2017. Semantics derived automatically from language corpora contain human-like biases. *Science*, 356(6334):183–186.

Myra Cheng, Esin Durmus, and Dan Jurafsky. 2023. Marked personas: Using natural language prompts to measure stereotypes in language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1504–1532, Toronto, Canada. Association for Computational Linguistics.

Maria De-Arteaga, Alexey Romanov, Hanna Wallach, Jennifer Chayes, Christian Borgs, Alexandra Chouldechova, Sahin Geyik, Krishnaram Kenthapadi, and Adam Tauman Kalai. 2019. Bias in bios: A case study of semantic representation bias in a high-stakes setting. In *proceedings of the Conference on Fairness, Accountability, and Transparency*, pages 120–128.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the*

North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers).

Samuel Gehman, Suchin Gururangan, Maarten Sap, Yejin Choi, and Noah A. Smith. 2020. RealToxicityPrompts: Evaluating neural toxic degeneration in language models. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 3356–3369, Online. Association for Computational Linguistics.

Hila Gonen and Yoav Goldberg. 2019. Lipstick on a pig: Debiasing methods cover up systematic gender biases in word embeddings but do not remove them. In *Proceedings of NAACL-HLT*.

Hila Gonen, Srini Iyer, Terra Blevins, Noah A Smith, and Luke Zettlemoyer. 2023. Demystifying prompts in language models via perplexity estimation. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 10136–10148.

Hila Gonen and Kellie Webster. 2020. Automatically identifying gender issues in machine translation using perturbations. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 1991–1995.

Thilo Hagendorff, Sarah Fabi, and Michal Kosinski. 2023. Human-like intuitive behavior and reasoning biases emerged in large language models but disappeared in chatgpt. *Nature Computational Science*, 3(10):833–838.

Daphne Ippolito, Ann Yuan, Andy Coenen, and Sehmon Burnam. 2022. Creative writing with an ai-powered writing assistant: Perspectives from professional writers. *arXiv preprint arXiv:2211.05030*.

Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38.

Erik Jones and Jacob Steinhardt. 2022. Capturing failures of large language models via human cognitive biases. *Advances in Neural Information Processing Systems*, 35:11785–11799.

Svetlana Kiritchenko and Saif Mohammad. 2018. Examining gender and race bias in two hundred sentiment analysis systems. In *Proceedings of the Seventh Joint Conference on Lexical and Computational Semantics*, pages 43–53, New Orleans, Louisiana. Association for Computational Linguistics.

Hannah Rose Kirk, Yennie Jun, Filippo Volpin, Haider Iqbal, Elias Benussi, Frederic Dreyer, Aleksandar Shtedritski, and Yuki Asano. 2021. Bias out-of-the-box: An empirical analysis of intersectional occupational biases in popular generative language models. *Advances in neural information processing systems*, 34:2611–2624.

Hadas Kotek, Rikker Dockum, and David Sun. 2023. Gender bias and stereotypes in large language models. In *Proceedings of the ACM collective intelligence conference*, pages 12–24.

Evelina Leivada, Elliot Murphy, and Gary Marcus. 2023. Dall· e 2 fails to reliably capture common syntactic processes. *Social Sciences & Humanities Open*, 8(1):100648.

Olivia Macmillan-Scott and Mirco Musolesi. 2024. (ir) rationality and cognitive biases in large language models. *Royal Society Open Science*, 11(6):240255.

Rowan Hall Maudslay, Hila Gonen, Ryan Cotterell, and Simone Teufel. 2019. It's all in the name: Mitigating gender bias with name-based counterfactual data substitution. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 5267–5275, Hong Kong, China. Association for Computational Linguistics.

David E Meyer and Roger W Schvaneveldt. 1971. Facilitation in recognizing pairs of words: evidence of a dependence between retrieval operations. *Journal of experimental psychology*, 90(2):227.

Moin Nadeem, Anna Bethke, and Siva Reddy. 2021. Stereoset: Measuring stereotypical bias in pretrained language models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 5356–5371.

Roberto Navigli, Simone Conia, and Björn Ross. 2023. Biases in large language models: origins, inventory, and discussion. *ACM Journal of Data and Information Quality*.

James H Neely. 1976. Semantic priming and retrieval from lexical memory: Evidence for facilitatory and inhibitory processes. *Memory & cognition*, 4(5):648–654.

Debora Nozza, Federico Bianchi, and Dirk Hovy. 2021. HONEST: Measuring hurtful sentence completion in language models. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 2398–2406, Online. Association for Computational Linguistics.

Aditya Ramesh, Prafulla Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. 2022. Hierarchical text-conditional image generation with clip latents. *arXiv preprint arXiv:2204.06125*.

Royi Rassin, Eran Hirsch, Daniel Glickman, Shauli Ravfogel, Yoav Goldberg, and Gal Chechik. 2024. Linguistic binding in diffusion models: Enhancing attribute correspondence through attention map alignment. *Advances in Neural Information Processing Systems*, 36.

Royi Rassin, Shauli Ravfogel, and Yoav Goldberg. 2022. Dalle-2 is seeing double: Flaws in word-to-concept mapping in text2image models. In *Proceedings of the Fifth BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, pages 335–345.

Nils Reimers and Iryna Gurevych. 2019. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*.

Paul Röttger, Hannah Kirk, Bertie Vidgen, Giuseppe Attanasio, Federico Bianchi, and Dirk Hovy. 2024. XSTest: A test suite for identifying exaggerated safety behaviours in large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 5377–5400, Mexico City, Mexico. Association for Computational Linguistics.

Teven Le Scao, Angela Fan, Christopher Akiki, Ellie Pavlick, Suzana Ilić, Daniel Hesslow, Roman Castagné, Alexandra Sasha Luccioni, François Yvon, et al. 2022. Bloom: A 176b-parameter open-access multilingual language model. *arXiv preprint arXiv:2211.05100*.

Timo Schick, Sahana Udupa, and Hinrich Schütze. 2021. Self-diagnosis and self-debiasing: A proposal for reducing corpus-based bias in nlp. *Transactions of the Association for Computational Linguistics*, 9:1408–1424.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2024. Quantifying language models' sensitivity to spurious features in prompt design or: How i learned to start worrying about prompt formatting. In *Proceedings of International Conference on Learning Representations*.

Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R Johnston, et al. 2024. Towards understanding sycophancy in language models. In *Proceedings of International Conference on Learning Representations*.

Gabriel Stanovsky, Noah A. Smith, and Luke Zettlemoyer. 2019. Evaluating gender bias in machine translation. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 1679–1684, Florence, Italy. Association for Computational Linguistics.

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. 2023. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*.

Endel Tulving and Daniel L Schacter. 1990. Priming and human memory systems. *Science*, 247(4940):301–306.

Endel Tulving, Daniel L Schacter, and Heather A Stark. 1982. Priming effects in word-fragment completion are independent of recognition memory. *Journal of experimental psychology: learning, memory, and cognition*, 8(4):336.

Pranav Narayanan Venkit, Sanjana Gautam, Ruchi Panchanadikar, Ting-Hao'Kenneth' Huang, and Shomir Wilson. 2023. Nationality bias in text generation. *arXiv preprint arXiv:2302.02463*.

Pranav Narayanan Venkit, Mukund Srinath, and Shomir Wilson. 2022. A study of implicit bias in pretrained language models against people with disabilities. In *Proceedings of the 29th International Conference on Computational Linguistics*, pages 1324–1332, Gyeongju, Republic of Korea. International Committee on Computational Linguistics.

Eric Wallace, Shi Feng, Nikhil Kandpal, Matt Gardner, and Sameer Singh. 2019. Universal adversarial triggers for attacking and analyzing NLP. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2153–2162, Hong Kong, China. Association for Computational Linguistics.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, et al. 2019. Huggingface's transformers: State-of-the-art natural language processing. *arXiv preprint arXiv:1910.03771*.

Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q Weinberger, and Yoav Artzi. 2020. Bertscore: Evaluating text generation with bert. In *Proceedings of International Conference on Learning Representations*.

Yuji Zhang, Sha Li, Jiateng Liu, Pengfei Yu, Yi R Fung, Jing Li, Manling Li, and Heng Ji. 2024. Knowledge overshadowing causes amalgamated hallucination in large language models. *arXiv preprint arXiv:2407.08039*.

Jieyu Zhao, Tianlu Wang, Mark Yatskar, Vicente Ordonez, and Kai-Wei Chang. 2018. Gender bias in coreference resolution: Evaluation and debiasing methods. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, pages 15–20, New Orleans, Louisiana. Association for Computational Linguistics.